

A Note on a Claimed Disproof of the Collatz Conjecture

Mary Lafontaine¹ Gerard Cheong²

Abstract

This note provides a critical analysis of a recent manuscript claiming a disproof of the Collatz conjecture. We demonstrate that the central step of the argument is not mathematically well-defined, due to a misuse of asymptotic notation. Consequently, the reasoning cannot support the claimed conclusion.

More generally, this note highlights the importance of exact use of asymptotic notation in discrete dynamical systems.

Keywords: Collatz conjecture; Big-O notation; asymptotic notation; misuse of notation; discrete dynamical systems.

1 Introduction

The Collatz conjecture remains one of the best-known open problems in elementary number theory. In May 2025, Mirzakhmet Syzdykov circulated a short preprint entitled *Disproof of Collatz Conjecture by using O-notation* [1], claiming a full disproof of the conjecture by means of Big-O notation.

Because such a claim concerns a longstanding open problem, and because the central argument relies on asymptotic notation in a nonstandard way, it is useful to examine the logical status of the proposed reasoning. The purpose of the present note is not to discuss the Collatz conjecture in general, but only to clarify why the key step of this claimed disproof is not mathematically well-defined.

The present note focuses exclusively on the central step of the claimed disproof. It does not attempt to provide a comprehensive analysis of all aspects of the manuscript.

1.1 The Collatz conjecture

The Collatz function is defined by

$$T(n) = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ 3n + 1 & \text{if } n \text{ is odd.} \end{cases}$$

The Collatz conjecture states that for every integer $n > 0$, the sequence obtained by iterating T eventually reaches 1.

1.2 Context and related work

The Collatz conjecture has generated an heterogeneous literature, ranging from rigorous mathematical analyses to numerous claimed proofs or disproofs.

A number of recent and classical works related to the Collatz problem may be found in [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18]. These contribution differ widely in scope and level of rigor, and illustrate the diversity of approaches that have been proposed.

In this context, it is important to clearly distinguish between heuristic reasoning and mathematically well-defined arguments. The present note focuses on a specific example in which a misuse of asymptotic notation leads to an invalid conclusion.

2 A Claimed Disproof

Syzdykov [1] proposes an argument intended to disprove the Collatz conjecture using Big-O notation applied to the length of Collatz trajectories.

The central step of the argument involves an inequality of the form

$$O(f(n)) \leq n.$$

This expression suffers from several fundamental issues.

First, Big-O notation does not represent a numerical value; rather, it denotes a class of functions. Writing $O(f(n))$ as if it were a function or a scalar quantity constitutes a category error.

Second, as a consequence, the expression $O(f(n))$ cannot be compared to an integer. Inequalities of the form $O(f(n)) \leq n$ are therefore not mathematically well-defined.

Formally, the statement

$$g(n) \in O(f(n))$$

means that there exist constants $C > 0$ and n_0 such that

$$|g(n)| \leq C|f(n)|$$

for all $n \geq n_0$. This definition does not assign a specific value to $O(f(n))$, but instead describes a set of functions.

Third, asymptotic notation characterizes the growth of functions as $n \rightarrow \infty$, and does not provide information about the exact value of a function for a given integer. Consequently, it cannot be used to derive pointwise bounds on the length of Collatz trajectories.

Since the central step of the argument relies on an expression that is not mathematically well-defined and on an incorrect use of asymptotic notation, the reasoning does not establish a valid disproof of the Collatz conjecture.

2.1 A formal clarification on category misuse

The expression

$$O(f(n)) \leq n$$

fails at a more fundamental level than an incorrect estimate: it is ill-typed.

Indeed, Big-O notation defines a set of functions,

$$O(f(n)) = \{g : \exists C > 0, \exists n_0, \forall n \geq n_0, |g(n)| \leq C|f(n)|\}.$$

Thus $O(f(n))$ is a mathematical object of set-theoretic nature, whereas n is a scalar integer.

Comparisons such as

$$O(f(n)) \leq n$$

attempt to order objects belonging to incompatible categories. They are therefore not false in the usual mathematical sense, but undefined.

This distinction is important: an incorrect inequality may still be meaningful and refutable, whereas an undefined expression cannot serve as a premise in a proof.

3 Discussion and methodological implications

The issue identified in the previous section is directly related to [1], but it reflects a more general pattern that regularly appears in discussions of the Collatz conjecture.

The problem often attracts arguments based on misuse or non-standard interpretation of familiar mathematical tools, in this context, asymptotic notation.

A recurrent source of confusion is the implicit treatment of asymptotic classes as numerical quantities. However, statements of the form

$$O(f(n)) \leq n$$

have no intrinsic meaning unless a specific function is chosen within the class $O(f(n))$. Without this specification, the comparison is not well-defined.

More generally, asymptotic bounds describe eventual growth behavior and do not imply pointwise inequalities. Even if a function $L(n)$ satisfies $L(n) \in O(f(n))$, this only guarantees the existence of constants $C > 0$ and n_0 such that $L(n) \leq Cf(n)$ for $n \geq n_0$. It does not justify conclusions about the exact value of $L(n)$ for individual integers.

This distinction is essential in discrete dynamical systems such as the Collatz iteration, where global growth properties and local stepwise behavior have different roles.

The present example illustrates how a seemingly minor ambiguity in notation can invalidate an entire argument.

Such issues are not specific to the Collatz problem and can be observed more generally in arguments where asymptotic reasoning is used to derive pointwise conclusions.

4 Conclusion

The argument proposed in [1] relies on an error of usage of the asymptotic notation and therefore does not constitute a valid mathematical disproof of the Collatz conjecture.

Such misunderstandings highlight the importance of rigorous use of asymptotic notation in number-theoretic contexts.

This note is the first in a series examining recent manuscripts related to the Collatz conjecture and establishing recurring patterns.

A number of recent and classical contributions related to the Collatz problem may be found in [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18]. In the present series of notes we intend to examine these works and related approaches in more detail.

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